Leave-one-out validation for generalized linear models

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1 Problem and motivation

Cross-validation evaluates predictive performance by testing a model on observations excluded from its training data. This helps assess how a model specification or fitting procedure predicts new subjects from the same population. A simple approach randomly splits the data into training and testing sets in a 2:1 ratio. However, the result depends on the random split, and using fewer observations for training reduces information available for estimation and can reduce statistical power. In K-fold cross-validation, K is the number of approximately equal groups. Each group is tested once after fitting on the other $K-1$ groups. Ten folds are commonly used. When evaluating a procedure that selects variables or models, selection must also occur within each training fold. Cross-validation can therefore assess an entire modeling procedure, as well as a fixed model specification. Leave-one-out validation (LOO) uses $K = n$, where n is the number of subjects. It fits on $n-1$ subjects and predicts the omitted subject in each round. This preserves almost all observations for training and is useful for small datasets, although it requires n fits.

When evaluating predictive performance, two simple statistics based on point predictions are mean absolute error (MAE) and root mean square error (RMSE). Let i index subjects from 1 to n, let y_i be subject i's observed outcome, and let $\widehat{y}_i$ be the prediction from a model fitted without that subject. Define the prediction error as $e_i = \widehat{y}_i - y_i$. Then

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|e_i|, \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}e_i^2}.$$

Both metrics use the outcome's scale, and smaller values indicate better predictions on the same evaluation data. RMSE gives greater weight to large errors. For a binary outcome, the prediction is an estimated event probability and the observed outcome is zero or one.

The ultimate goal of this project is to develop an R package that makes it easier to perform LOO validation for generalized linear models (GLMs) and inspect prediction metrics. At this midterm stage, the core functions have been implemented. They automate model fitting, prediction and error calculation and return an object that users can print, summarize and plot.

2 Using the functions

The workflow has three core functions. In each LOO iteration, `fit_fun()` fits one model using all subjects except the omitted subject. After all iterations, `predict_fun()` uses the collected coefficients to calculate predictions, individual errors, MAE and RMSE. `run_loo()` coordinates the iterations and calls both functions.

```
source("../R/main.R")
```

Function	Main inputs	Output
`fit_fun()`	`formula`, training `data`, `family`, `link`	Named vector of fitted coefficients
`predict_fun()`	`beta`, full dataset `dat`, `formula`, `family`, `link`	A `loo_results` object
`run_loo()`	`formula`, full dataset `data`, `family`, `link`, `workers`	A `loo_results` object

For `fit_fun()`, `formula` specifies the response and predictors, and `data` supplies the training observations for that fit. `family` is a function such as `gaussian` or `binomial`, and `link` is a string such as `"identity"` or `"logit"`.

The returned S3 object has the following contents. S3 provides methods selected according to an object's class.

Component	Meaning
`formula`, `family`	Model formula and family specification, including the link
`nrow`, `ncol`	Numbers of rows and columns in the input data frame
`coefficients`	Matrix of omitted-row coefficient estimates
`observed`, `predicted`	Observed responses and corresponding LOO predictions
`bias`	Individual errors, computed as predicted minus observed
`mae`, `rmse`	MAE and RMSE across all subjects

For an object named `result`, the three S3 methods are:

- `print(result)` displays MAE and RMSE.
- `summary(result)` displays the formula, dimensions, family, link and metrics.
- `plot(result, bins = 10L)` returns a ggplot error histogram. `bins` is the number of bins.

3 Simulation example and results

The sourced script uses random seed 648 to generate two datasets with the same $n = 50$ subjects and the same five predictors. In the table below, x_{ji} is the value of predictor j for subject i, where j ranges from 1 to 5. SD denotes standard deviation.

Predictor	Distribution	Parameters
x_{1i}	Normal	Mean 0, SD 1
x_{2i}	Normal	Mean 5, SD 5
x_{3i}	Poisson	Rate 3
x_{4i}	Bernoulli	Probability 0.3
x_{5i}	Uniform	Minimum 0, maximum 10

Predictors are generated independently across variables and subjects. Define $z_i = (1, x_{1i}, x_{2i}, x_{3i}, x_{4i}, x_{5i})^\mathsf{T}$ as the predictor vector including an intercept. The continuous and binary coefficient vectors are

$$\beta_c = (1, 1, -0.5, 2, -2, 0.5)^\mathsf{T},$$
$$\beta_b = (-1, 0.3, -0.1, 0.3, -0.5, 0.1)^\mathsf{T}.$$

The subscripts c and b identify the continuous and binary models. The continuous response is $Y_{ci} = z_i^\mathsf{T}\beta_c + \varepsilon_i$, where $\varepsilon_i \sim N(0, 1)$ is independent noise. The binary response is $Y_{bi} \sim \text{Bernoulli}(p_i)$, with true event probability $p_i = \text{logit}^{-1}(z_i^\mathsf{T}\beta_b)$. The corresponding data frames are `cont` and `binary`, with responses stored in `Y`.

To use `run_loo()`, supply a model formula, a data frame, a family function and a link. The default `workers = 1L` runs serially, and a larger positive integer enables parallel execution. The formula below automatically includes an intercept. These calls use a Gaussian family with identity link for the continuous response and a binomial family with logit link for the binary response.

```
model_formula <- Y ~ x1 + x2 + x3 + x4 + x5
cont_result <- run_loo(model_formula, cont, gaussian, "identity")
binary_result <- run_loo(model_formula, binary, binomial, "logit")
print(cont_result)
```

```
Leave-one-out prediction metrics
MAE:  0.857753
RMSE: 1.038675
```

```
summary(binary_result)
```

```
Formula: Y ~ x1 + x2 + x3 + x4 + x5
Data: 50 rows and 7 columns
Family: binomial | Link: logit
Leave-one-out prediction metrics
MAE:  0.501916
RMSE: 0.535407
```

The continuous model has MAE 0.858 and RMSE 1.039. The binary model has MAE 0.502 and RMSE 0.535. These binary scores compare estimated probabilities with realized zero or one outcomes. They can remain substantial even when the probability model is correctly specified.

The plotting method displays the distribution of the individual LOO errors.

```
plot(binary_result)
```

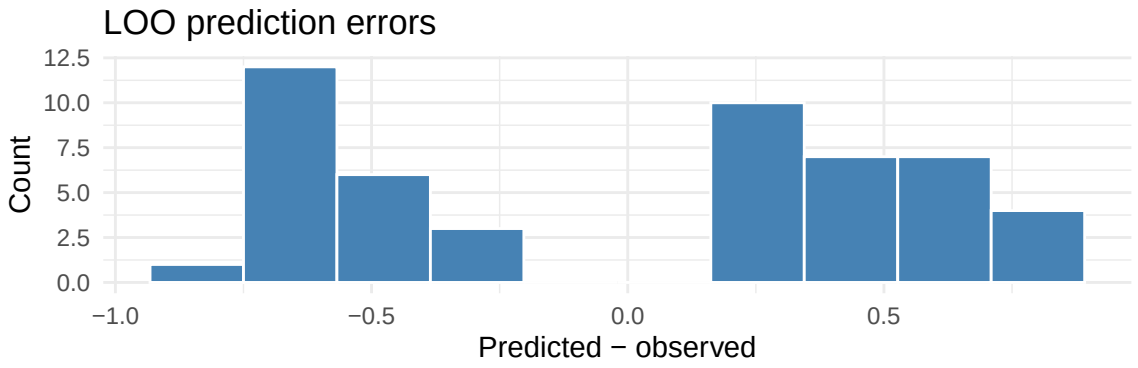


Figure 1: Binary LOO prediction errors, defined as predicted probability minus observed outcome.

AI disclosure. AI assisted with coding based on my pseudocode. AI also helped format the project in LaTeX, while I wrote and revised the content.